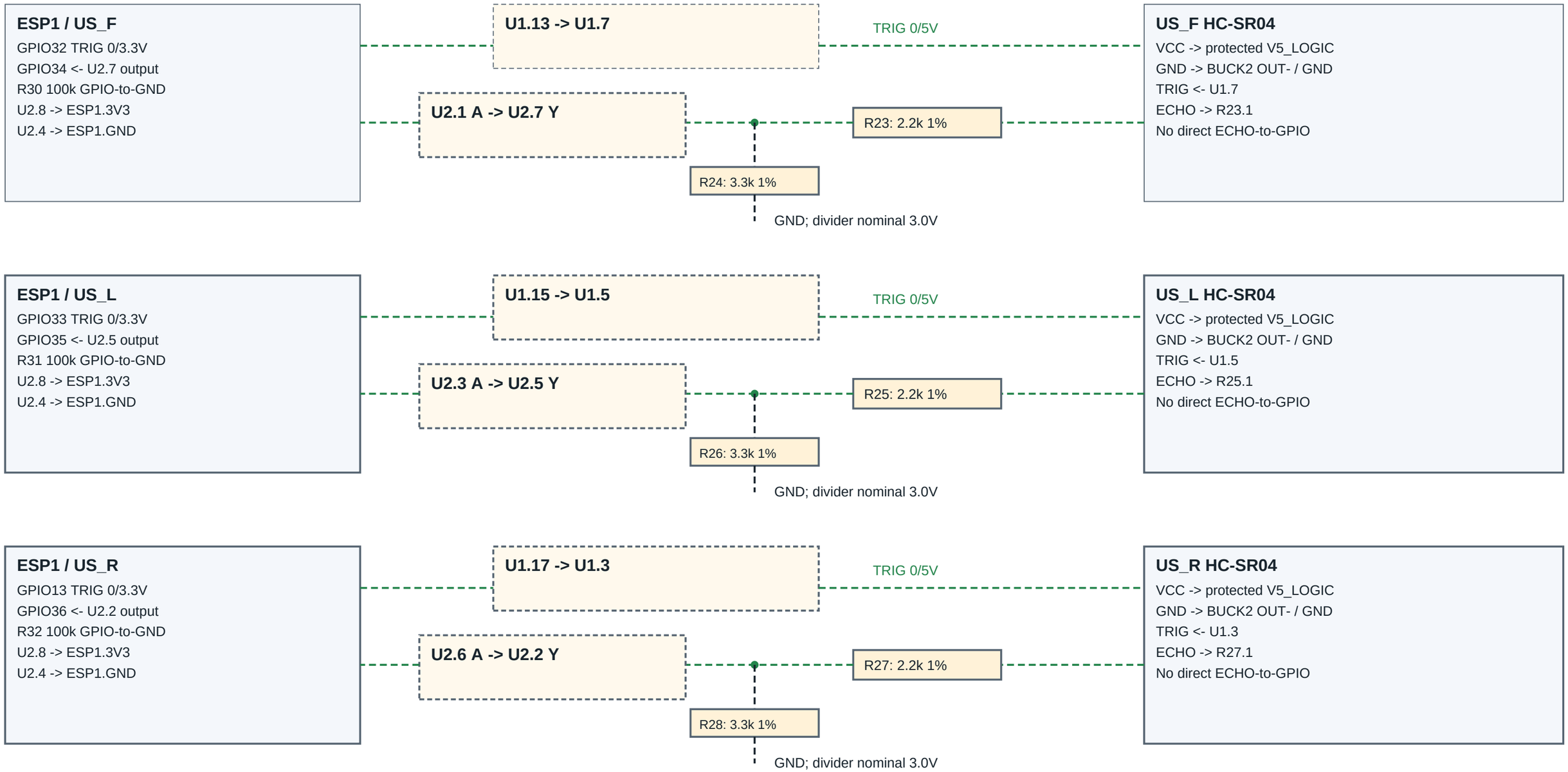
HC-SR04: dividers plus powered-off ECHO isolation

U2 = SN74LVC3G34DCU; supply from MCU 3V3, not sensor 5V. Same GPIOs. Verify actual pulse levels/edges before connection.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026



C_U2: 100nF U2.8-to-U2.4. U2 inputs: HIGH >=2.0V, LOW <=0.8V; edge rate <=10ns/V at 3.3V +/-0.3V. Keep divider-to-U2 traces short. Scope actual sensor cable and pulse shape. U2 outputs follow MCU supply; power-off Ioff limits backfeed.

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.